

Year 7 Computer Science

Practical Learning Checkpoint

TURTLE RESCUE

Collect the power cells and find the safe exit.

Student name: _____

Class: _____ **Date:** _____

Duration: 40 minutes

Core tasks: 20 marks

YOUR MISSION

A robot needs two power cells, A and B. Then it must go around a blocked room to reach EXIT.
Write two short Python Turtle programs. They draw route lines, not a complete game.

WAGBA: Predict, run, fix and extend Turtle code to reach a target safely.

Knowledge: x is left/right; y is up/down. A turn changes direction.

Skills: Change a command, run the program and check its drawing.

Understanding: The numbers and order of commands change the route.

How to work

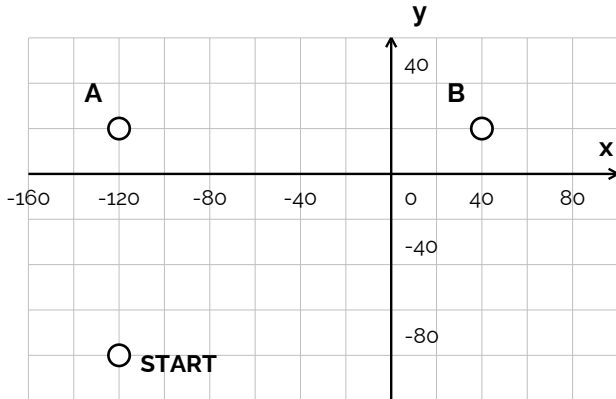
1. Use the class Turtle editor, this paper and its command reminders.
2. You may discuss ideas with a partner or group. Run your own program and record your own choices and final commands.
3. Do Predict before Run. An incorrect attempt does not stop you: check, change one command and try again.
4. Do both core tasks on pages 2-5. If time remains, choose an extra challenge on pages 6-8. Extra challenges have no marks.
5. No screenshots or long explanations are required. Your teacher will check your code and drawing. Save if your editor allows it.

Timing: 3 min mission | 12 min Task 1 | 16 min Task 2 | 6 min finish/extra | 3 min check

Task 1 - Collect A, then B

Use this map. Your Turtle output only needs to show the route line. [10 marks total]

TARGET MAP



START = (-120, -80)

A = (-120, 20)

B = (40, 20)

STARTER CODE

```
1 import turtle as t
2
3 t.penup()
4 t.goto(-120, -80)
5 t.pendown()
6
7 t.goto(-120, -20)
8
9 t.done()
```

x = left / right

y = up / down

Each grid square = 20 units.

BEFORE YOU START

Use the starter code in a fresh program. Do not type the line numbers. It starts at START without drawing from the centre. You only need the route line, not the power cells or grid.

01.1 PREDICT

[1 mark]

Read line 7. Where will the Turtle stop? Tick one before running.

☐ (-120, -20)

☐ (-120, 20)

☐ (40, 20)

01.2 RUN

[1 mark]

Press Run. Look at the end of the line. Did it reach A? Tick one.

☐ Yes

☐ Not yet

01.3 INVESTIGATE

[1 mark]

Compare line 7 with A = (-120, 20). Which number must change?

☐ The x number

☐ The y number

Next: turn to page 3 to fix the line and continue to B.

Task 1 - Fix and continue

Goal: one continuous line from START to A, then from A to B.

01.4 MODIFY

[3 marks]

Change only the y number in line 7 so the line reaches A. Run again.

CHECK YOUR DRAWING

A is (-120, 20). The first line should go straight up from START to A. Keep the x number unchanged. You can try again if the line stops in the wrong place.

☐ I ran my changed program and checked its endpoint.

01.5 MAKE

[4 marks]

Add one new `t.goto` command to reach B = (40, 20). Put it after the command for A and before `t.done()`. Run the whole route.

COMMAND REMINDER

`t.goto(x, y)` moves to a position. The first number is x; the second is y. Keep the pen down so the route is a continuous line. Keep your working command for A.

Copy your TWO successful route commands below. Do not copy the setup or `t.done()`.

Your successful drawing commands

Final check

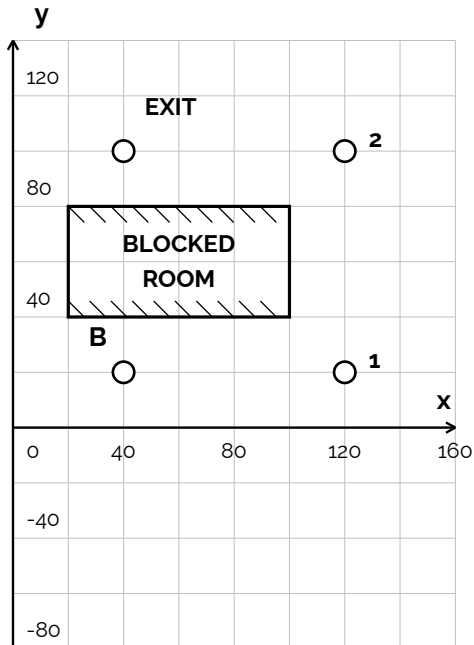
- ☐ The line starts at START, not at the centre.
- ☐ It reaches A first, then B.
- ☐ I ran the final program and checked the two line sections.

Save as TurtleRescue_Task1_v1.py if possible. Otherwise keep your commands on this paper and show the drawing to your teacher. Continue to page 4.

Task 2 - Find the safe exit

Start at B. Go around the blocked room, using corners 1 and 2. [10 marks total]

TARGET MAP



STARTER CODE

```
1 import turtle as t
2
3 t.penup()
4 t.goto(40, 20)
5 t.pendown()
6
7 t.forward(80)
8 t.right(90)
9 t.forward(80)
10
11 t.done()
```

Fresh Turtle faces right.

x: left/right. y: up/down.

B = (40, 20)

EXIT = (40, 100)

Corner 1 = (120, 20)

Corner 2 = (120, 100)

YOUR JOB

Open a fresh program with this starter code. The route must travel right, then up, then left to EXIT. Keep the line outside the blocked room. The starter has a wrong turn.

02.1 PREDICT

[1 mark]

After `t.right(90)`, will the next line go up or down? Tick one.

☐ Up

☐ Down

02.2 RUN

[1 mark]

Run the starter. Which way did the second line go? Tick one.

☐ Up

☐ Down

02.3 INVESTIGATE

[1 mark]

Which command changes the direction? Tick one.

☐ `t.forward(80)`

☐ `t.right(90)`

☐ `t.pendown()`

Task 2 - Repair the turn and escape

Goal: B -> corner 1 -> corner 2 -> EXIT. The line must not cross the blocked room.

02.4 MODIFY

[3 marks]

Replace the wrong turn so the second line goes UP to corner 2. Keep both distances at 80. Run and check.

LOOK AT THE MOVEMENT

`t.forward(80)` moves 80 units in the direction the Turtle faces. `t.left(90)` and `t.right(90)` turn the Turtle without moving it. Check where it faces before the next move.

☐ I ran the repaired program and reached corner 2 = (120, 100).

02.5 MAKE

[4 marks]

Continue LEFT along the top of the room to EXIT = (40, 100). Add a `t.goto` command OR a turn and a forward command before `t.done()`. Run the complete route.

Copy all your successful route commands below, starting with `t.forward(80)`. Do not copy the setup or `t.done()`.

Your successful drawing commands

Check and finish

- ☐ My route visits corner 1, then corner 2, then EXIT.
- ☐ The line stays outside the blocked room.
- ☐ I can point to one changed command and tell a partner what it does.

Save as `TurtleRescue_Task2_v1.py` if possible. Hand in this paper. If time remains, choose ONE extra challenge on pages 6-8.

Shape Studio 1 - Rectangle builder

Extra challenge - no marks. Choose one after finishing both core tasks.

YOUR NEW GOAL
Turn the sample square into a rectangle 80 units wide and 40 units tall. Then move the whole drawing to the right side of the screen.

```
1 import turtle as t
2 t.penup()
3 t.goto(-140, -60)
4 t.pendown()
5 t.forward(60)
6 t.right(90)
7 t.forward(60)
8 t.right(90)
9 t.forward(60)
10 t.right(90)
11 t.forward(60)
12 t.right(90)
13 t.done()
```

FOLLOW PRIMM

PREDICT
Read the sample. Which shape will it draw? Say your prediction to a partner.

RUN
Run the sample and look at its four equal sides.

INVESTIGATE
Find the four forward commands. The 1st and 3rd draw the width; the 2nd and 4th draw the height.

MODIFY
Use distances 80, 40, 80, 40. Keep all four turns at 90. Run again.

MAKE
Change the starting position to (20, -60). Run your rectangle on the right side of the screen.

Record only your eight movement/turn commands. Do not copy the setup. Starting position for your final drawing: (20, -60).

Your successful drawing commands

Your successful drawing commands

Write commands 1-4 in the left box, then 5-8 in the right box.

Shape Studio 2 - Triangle explorer

Extra challenge - no marks. Choose one after finishing both core tasks.

YOUR NEW GOAL

Make a larger triangle with three 90-unit sides. Then place it on the right of the centre. The sample uses 120-degree turns between sides.

```
1 import turtle as t
2 t.penup()
3 t.goto(-120, 0)
4 t.pendown()
5 t.forward(60)
6 t.left(120)
7 t.forward(60)
8 t.left(120)
9 t.forward(60)
10 t.left(120)
11 t.done()
```

FOLLOW PRIMM

PREDICT

How many sides will the sample draw? Tell a partner before running.

RUN

Run it. Check that the three sides join.

INVESTIGATE

Point to the three moves and the three turns. The 120-degree turn is how far the Turtle turns, not the triangle's inside angle.

MODIFY

Change every forward distance from 60 to 90. Keep each turn at 120. Run the larger triangle.

MAKE

Change the starting position to (40, 0). Run again. Compare its new position with the sample.

Copy the six successful movement/turn commands below. Do not copy the setup. Final starting position: (40, 0).

Your successful drawing commands

Shape Studio 3 - My own drawing

Extra challenge - no marks. Choose one after finishing both core tasks.

YOUR NEW GOAL

Create a robot face or a small house using at least six drawing commands. Your design should include at least two turns or changes of direction.

```
1 import turtle as t
2 t.penup()
3 t.goto(-100, -40)
4 t.pendown()
5 t.goto(-100, 40)
6 t.goto(-20, 40)
7 t.done()
```

FOLLOW PRIMM

PREDICT

Read the sample. Say where its two lines will go.

RUN

Run it to see an upside-down L on the left of the centre.

INVESTIGATE

Point to the start and the two destinations. The first coordinate in each pair is x.

MODIFY

Reflect the L to the right: change the x values -100 and -20 to 100 and 20. Keep y values the same.

MAKE

Start a fresh program. Use goto, forward, left or right to draw your own robot face or house. Run, change and improve it.

Sketch your own idea here:



Copy six successful drawing commands from your own design. Continue on the back if needed. Do not copy import, setup or t.done().

Your successful drawing commands
